

STORNOWAY, United Kingdom

WMO: 030260

Lat: 58.2136N

Lon: 6.3189W

Elev: 8

StdP: 101.23

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-1.2	-0.1	-4.7	2.5	1.4	-3.6	2.8	1.8	19.6	6.7	18.3	6.7	3.1	310	0.787

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.0	18.8	15.5	17.3	14.6	16.3	14.0	16.2	17.9	15.2	16.6	14.5	15.8	4.8	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
15.3	10.9	17.1	14.5	10.3	16.0	13.8	9.8	15.2	45.3	18.1	42.7	16.7	40.7	15.8	22.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.4	14.4	12.8	DB	-4.1	22.0	2.8	2.0	-6.1	23.4	-7.8	24.6	-9.4	25.7	-11.5	27.1	(4)
(5)			WB	-4.5	17.9	1.7	1.5	-5.7	19.0	-6.7	19.9	-7.6	20.7	-8.9	21.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.8	5.1	5.0	5.7	7.4	9.2	11.6	13.4	13.5	12.0	9.6	7.1	5.4		
	DBStd	3.83	2.30	2.38	2.27	2.31	2.30	1.92	1.87	1.70	1.93	2.29	2.56	2.83		
	HDD10.0	832	152	141	133	83	42	6	1	0	5	36	91	144		
	HDD18.3	3494	411	374	391	329	282	203	153	151	191	271	338	401		
	CDD10.0	381	0	1	1	4	18	53	106	108	64	23	3	1		
	CDD18.3	1	0	0	0	0	0	0	0	1	0	0	0	0		
	CDH23.3	1	0	0	0	0	0	0	1	0	0	0	0	0		
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0		
(14) Wind	WSAvg	6.3	7.3	7.3	7.1	6.2	5.7	5.7	5.1	5.2	5.7	6.5	6.6	6.9		
(15) Precipitation	PrecAvg	1191	132	97	102	70	64	64	73	85	109	132	130	130		
	PrecMax	1547	269	271	206	116	153	112	164	203	206	231	228	268		
	PrecMin	800	26	10	19	1	13	15	23	16	34	38	38	23		
	PrecStd	162	54	55	44	25	30	23	29	40	41	48	43	54		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	11.4	11.9	15.3	17.9	19.3	21.2	20.7	18.1	15.5	13.6	12.3		
		MCWB	10.3	10.1	9.3	11.0	13.7	15.2	17.4	16.8	14.5	13.3	12.0	11.1		
	2%	DB	10.3	10.5	10.7	12.8	15.2	16.7	19.0	18.3	16.4	14.4	12.3	11.4		
		MCWB	9.4	9.5	9.0	9.8	11.9	13.9	16.0	15.4	13.9	12.6	11.0	10.4		
	5%	DB	9.6	9.7	9.8	11.6	13.7	15.3	17.4	16.9	15.5	13.7	11.6	10.6		
		MCWB	8.7	8.7	8.4	9.2	11.0	12.8	14.9	14.6	13.6	12.2	10.4	9.6		
	10%	DB	8.8	8.8	9.1	10.6	12.6	14.3	16.2	16.0	14.8	13.0	10.9	9.8		
		MCWB	7.8	7.7	7.9	8.7	10.3	12.1	14.1	14.0	13.1	11.6	9.8	8.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.6	10.6	10.5	11.4	14.2	15.7	17.9	17.4	15.5	13.9	12.5	11.6		
		MCDB	10.9	11.0	11.3	13.8	17.3	18.6	20.6	20.0	16.7	14.7	13.1	11.9		
	2%	WB	9.7	9.7	9.5	10.5	12.5	14.2	16.4	16.0	14.7	13.2	11.5	10.7		
		MCDB	10.1	10.2	10.2	12.1	14.5	16.0	18.2	17.6	15.6	14.0	12.1	11.2		
	5%	WB	8.9	8.9	8.8	9.7	11.5	13.3	15.4	15.1	14.1	12.5	10.6	9.8		
		MCDB	9.5	9.6	9.6	11.1	13.2	14.7	16.9	16.4	15.1	13.4	11.4	10.5		
	10%	WB	7.9	7.8	8.1	9.1	10.7	12.6	14.5	14.4	13.5	11.8	9.9	8.9		
		MCDB	8.7	8.6	9.0	10.4	12.2	13.9	15.8	15.6	14.5	12.8	10.8	9.8		
(35) Mean Daily Temperature Range	5% DB	MDBR	4.1	4.4	4.8	5.4	5.7	4.9	5.0	5.0	4.9	4.7	4.4	4.5		
		MCDBR	4.5	4.5	5.1	6.3	7.0	6.3	6.7	6.0	5.4	4.7	4.5	4.7		
	5% WB	MCWBR	4.6	4.3	4.2	4.1	4.4	4.0	4.2	3.8	3.8	3.9	4.5	5.0		
		MCWBR	4.6	4.4	4.7	5.7	6.6	5.7	6.2	5.4	4.8	4.4	4.7	4.9		
(40) Clear-Sky Solar Irradiance	taub		0.353	0.356	0.370	0.383	0.382	0.373	0.377	0.369	0.365	0.350	0.329	0.320		
		taud	2.109	2.240	2.298	2.318	2.378	2.444	2.446	2.477	2.452	2.360	2.204	2.100		
	Edn at Noon		437	655	767	824	852	865	852	834	774	662	473	354		
		Edh at Noon	56	77	94	108	109	103	101	92	81	67	49	41		
(44) All-Sky Solar Radiation	RadAvg		0.33	0.91	2.03	3.57	4.75	4.80	4.44	3.58	2.35	1.19	0.46	0.22		
	RadStd		0.04	0.09	0.14	0.24	0.34	0.43	0.31	0.23	0.12	0.10	0.04	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	+1.01	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page